

Current Cybersecurity, Cryptography and Cyber Law — Detailed Lecture Notes

JKSSB Computer Awareness

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64.1 Scope and study method

This lesson covers the current-affairs edge of the official Computer Awareness scope: computer virus and latest anti-virus, terms and abbreviations used in IT, and the role of IT in governance. It treats cybersecurity threats, cryptography basics, firewall types, and cyber law at awareness level. The exam depth is recall and classification, with at most a one-step scenario. The study method for every term is the near-neighbour contrast: what it is, what it is not, and the nearest confusion listed alongside it.

Term	Meaning in this lesson
Malware	Hostile software, including virus, worm, Trojan horse, ransomware, spyware.
Parasitic / file-infector virus	A virus that attaches itself to an executable file.
Phishing	Fraudulent messages that trick the user into revealing credentials.
Ransomware	Malicious code that locks or encrypts files and demands payment.
DDoS	Distributed Denial of Service: flood of traffic from many machines.
Hacking	Unauthorised access to a computer or account.
DES — Data Encryption Standard	Symmetric cipher working on 64-bit blocks.
Encryption	Conversion of plaintext to ciphertext for confidentiality.
Digital signature	Authenticity and integrity proof; not confidentiality.
Firewall	Barrier between a trusted and an untrusted network.
Personal data	Data identifying a person, protected by consent and purpose limits.
Copyright	Owner's exclusive rights over an original work.

64.2 Malware and the parasitic virus

Malware is the general name for hostile software. A **virus** is malware that needs a host program and some human action, such as opening an attachment, to spread. A **worm** spreads on its own across a network without a host program. A **Trojan horse** looks like useful software but hides a harmful payload, and it does not self-replicate.

Definition (Parasitic (file-infector) virus). *A virus that attaches itself to an executable file. When the infected program runs, the virus code runs first and then hands control back to the host program, spreading further as infected files are shared or copied.*

The parasitic virus is the file-infecting member of the family: it lives inside executable files, activates when the host program runs, and travels with shared files. An **anti-virus** detects, blocks, and removes such infections, and its database must be kept updated against the latest strains.

Example 64.2.1. *A student copies a free game from a friend's pen drive and runs it on the office desktop. The game file carries a file infector: the virus runs first, attaches to other executable files, and spreads when those files are shared further. Updating the anti-virus and scanning the drive before opening the files is the correct defence.*

64.3 Phishing, ransomware, DDoS, and hacking

These four threats differ in mechanism, and the exam swaps their definitions. **Phishing** attacks the user: a fraudulent mail or message, often styled as a bank or KYC notice, tricks the user into revealing a password or OTP on a fake login page. **Ransomware** attacks the data: malicious code locks or encrypts the victim's files and demands payment for the decryption key. **DDoS** (Distributed Denial of Service) attacks availability: a flood of traffic from many machines overwhelms a server so genuine users cannot reach it. A flood from a single machine is only DoS; the distributed prefix matters. **Hacking** is the broad term for unauthorised access to a computer or account; phishing, ransomware, and DDoS are specific methods or goals within that broad idea.

Definition (Ransomware). *Malicious code that locks or encrypts the victim's files and demands payment for their release.*

Example 64.3.1. *An item describes a clerk whose files are suddenly unreadable, with a message demanding payment for the key. Tracing the mechanism — files locked, payment demanded — gives ransomware. If the stem instead described a fake bank mail asking for an OTP, the same trace gives phishing; a server slowed to a halt under flood traffic gives DDoS.*

64.4 Cryptography and DES

Cryptography protects information by transforming it so only the intended recipient can read it. **Plaintext** is the readable message, **ciphertext** is the transformed message, and a **key** controls the transformation. **Encryption** converts plaintext to ciphertext and **decryption** converts it back. In **symmetric** cryptography the same key encrypts and decrypts; in **asymmetric** cryptography a public key and a private key form a pair.

Definition (DES (Data Encryption Standard)). *A symmetric cipher that processes data in 64-bit blocks. Its key is 64 bits long, of which 56 bits are effective, the remaining 8 bits being parity.*

The two numbers for DES must be kept apart. The **block size** is 64 bits: that is the chunk of data processed at a time. The **effective key** is 56 bits. An option offering 128 bits confuses DES with a later cipher's block or key size.

64.5 Encryption vs digital signature

This is the TRAP-15 distinction and it recurs in papers. **Encryption** provides **confidentiality**: only the intended recipient can read the message. A **digital signature** provides **authenticity**

and integrity: it proves who sent the message and that it was not altered in transit. A signature does not hide the message content, and encryption by itself does not prove the sender's identity.

Definition (Digital signature). *A cryptographic proof attached to a message that establishes the sender's authenticity and the message's integrity, without by itself providing confidentiality.*

Example 64.5.1. *An item states, "A digital signature encrypts the message body so only the recipient can read it." This is wrong. Hiding the body is encryption (confidentiality). The signature answers two different questions: who sent it and whether it was changed. The wrong option has merged the two security verbs.*

64.6 Firewalls: three types

A **firewall** is a barrier between a trusted internal network and an untrusted external network. It permits or denies traffic according to a rule set. It filters packets and connections; it does not disinfect file-infecting viruses inside downloads that its rules allow through — that is the anti-virus's job.

A **packet-filtering** firewall examines each packet's headers — source and destination address, port, and protocol — and decides packet by packet. It is the fastest and simplest type but cannot judge a session as a whole. A **stateful-inspection** firewall examines the same headers plus the **connection state**: it remembers open sessions and drops packets that do not belong to a known session. An **application (proxy)** firewall sits between the client and the server as a go-between and inspects application-layer content and commands, such as web or mail requests.

Definition (Firewall). *A network barrier that permits or denies traffic between a trusted internal network and an untrusted external network according to a rule set.*

The ordering shorthand is: header only, then header plus state, then full content via proxy. Filtering checks the envelope, stateful inspection checks the conversation, and the proxy reads the letter inside.

64.7 Internet governance, privacy, and data protection

The Internet has no single owner; names and numbers are coordinated internationally, for example by ICANN (Internet Corporation for Assigned Names and Numbers). Governance concerns at awareness level are four: the **security** of networks, the **privacy** of personal data, fair **access** to the network, and which country's **law** applies when an act crosses borders. The positive side is the **role of IT in governance** (e-governance): online services, transparency, and delivery of benefits to citizens.

Personal data is any data that identifies a person, such as a name combined with a phone number, biometrics, or account details. Protection principles at awareness level are: collect data with **consent**, use it only for the stated **purpose**, keep it **secure**, and retain it no longer than needed. Privacy settings, strong passwords, and OTP discipline are the user-side defences. Misuse or leakage of personal data can attract legal consequences under cyber law.

64.8 Cyber law and copyright: awareness-level map

Cyber law at this level means two things: unauthorised access, fraud, and data misuse attract punishment, and creators hold enforceable rights over their works. The **Copyright Act** protects original literary, artistic, musical, and software works from copying without permission. The

awareness-level map matches each concept to its area: the **meaning of copyright** (what exclusive rights the owner holds), **first ownership** (who owns the work first), **infringement** (what counts as violation), **fair dealing** (limited permitted uses such as private study), and **offences** (punishment for wilful infringement). Software piracy — copying licensed software without permission — is the standard infringement example. Section numbers are recall only where the question itself supplies them; otherwise the concept match is what the exam tests.

64.9 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. DES block size is **64 bits**; the effective key is 56 bits. Do not offer 128 for either (DES fact).
2. Digital signature $\neq$ encryption: authenticity/integrity vs confidentiality (TRAP-15/TRM-11).
3. A file infector attaches to **executable files**; it is not a worm (self-spreading) or a Trojan (non-replicating disguise).
4. Packet filter checks headers only; stateful adds **connection state**; proxy inspects **application content**.
5. Phishing steals **credentials**; ransomware locks **files**; DDoS blocks **access**; hacking is unauthorised access generally.
6. DDoS needs **many** machines (distributed); a single-machine flood is only DoS.
7. Copyright protects the owner against **infringement**, with limited **fair-dealing** exceptions such as study — piracy is the standard violation example.

Example 64.9.1. *An item asks which firewall remembers open sessions and drops stray packets. Trace the triple: the packet filter decides per packet, the proxy inspects application content, and only stateful inspection tracks connection state. The answer is the stateful-inspection firewall. If the stem instead asked which device disinfects an infected executable, the answer leaves firewalls entirely and gives the anti-virus.*

Exercises

1. Define a parasitic (file-infector) virus and distinguish it from a worm and a Trojan horse.
2. State the DES block size and effective key size, and expand the abbreviation DES.
3. Distinguish packet-filtering, stateful-inspection, and application/proxy firewalls.
4. Distinguish phishing, ransomware, DDoS, and hacking in one line each.
5. Explain why a digital signature is not encryption, using TRAP-15.
6. List the four Internet-governance concerns and the four data-protection principles from this lesson.
7. Match each copyright concept (meaning, first ownership, infringement, fair dealing, offences) to what it covers.

Solutions

1. A parasitic virus attaches to an executable file, runs when the host program runs, and spreads with shared files. A worm spreads on its own across a network without a host; a Trojan horse disguises itself as useful software and does not self-replicate.
2. DES (Data Encryption Standard) is a symmetric cipher with a 64-bit block size and a 56-bit effective key (64 bits with parity).
3. Packet-filtering checks per-packet headers only; stateful inspection adds connection-state tracking; the application/proxy firewall inspects application-layer content as a go-between.
4. Phishing tricks the user into revealing credentials; ransomware locks or encrypts files for ransom; DDoS floods a service from many machines so genuine users cannot reach it; hacking is unauthorised access generally.
5. Encryption provides confidentiality (who can read it); a digital signature provides authenticity and integrity (who sent it, whether it was changed). Merging them is TRAP-15.
6. Governance concerns: security, privacy, access, jurisdiction (applicable law). Data-protection principles: consent, stated purpose, security, limited retention.
7. Meaning: the exclusive rights the owner holds; first ownership: who owns the work first; infringement: what counts as violation (for example software piracy); fair dealing: limited permitted uses such as private study; offences: punishment for wilful infringement.